

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - VI

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Class :BSC IT	Batch : 01
Date of Assignment :19/08/2026	Date/Time of Submission :19/08/2026

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.head())
```

```

      Roll no      Name Department  Marks  Attendance
0         39  Divyanshu     BSC IT     50           75
1         46   Abhishek     BSC CS    100           80
2         30    Pradu     BAMMC     75          100
3         36    Rahul     BSC IT     65           85
4         56     Aman     BSC CS     78           90

```

```
print(df.tail())
```

```

      Roll no      Name Department  Marks  Attendance
5         54    Rohit     BAMMC     82           88
6         44    Kunal     BSC IT     70           76
7         45   Vishal     BSC CS     92           95
8         14    Arjun     BAMMC     68           80
9          7     Sagar     BSC IT     88           92

```

```
print(df.shape) #number of rows and columns
```

```
(10, 5)
```

```
print(df.columns)
```

```
Index(['Roll no', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```

Roll no      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object

```

```
print(df.describe)
```

```

<bound method NDFrame.describe of
0         39  Divyanshu     BSC IT     50           75
1         46   Abhishek     BSC CS    100           80
2         30    Pradu     BAMMC     75          100
3         36    Rahul     BSC IT     65           85
4         56     Aman     BSC CS     78           90
5         54    Rohit     BAMMC     82           88
6         44    Kunal     BSC IT     70           76
7         45   Vishal     BSC CS     92           95

```

8	14	Arjun	BAMMC	68	80
9	7	Sagar	BSC IT	88	92>

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Attendance"]>=90]
print(result)
```

	Roll no	Name	Department	Marks	Attendance
2	30	Pradu	BAMMC	75	100
4	56	Aman	BSC CS	78	90
7	45	Vishal	BSC CS	92	95
9	7	Sagar	BSC IT	88	92

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

	Roll no	Name	Department	Marks	Attendance
0	39	Divyanshu	BSC IT	50	75
3	36	Rahul	BSC IT	65	85
8	14	Arjun	BAMMC	68	80
6	44	Kunal	BSC IT	70	76
2	30	Pradu	BAMMC	75	100
4	56	Aman	BSC CS	78	90
5	54	Rohit	BAMMC	82	88
9	7	Sagar	BSC IT	88	92
7	45	Vishal	BSC CS	92	95
1	46	Abhishek	BSC CS	100	80

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks ",df["Marks"].max())
print("Minimum Marks ",df["Marks"].min())
print("Average Marks ",df["Marks"].mean())
print("Total Marks ",df["Marks"].sum())
```

```
Maximum Marks 100
Minimum Marks 50
Average Marks 76.8
Total Marks 768
```